Technology Stack

Wandr AI – AI Travel Itinerary Planner

Frontend

Technology	Version	Purpose
React	18.3.x	UI component framework
TypeScript	~5.6.x	Static type safety
Vite	^5.4	Dev server, bundler, HMR
React Router DOM	^6.26	Client-side routing (SPA)
Tailwind CSS	^3.4	Utility-first styling
Supabase JS	^2.45	Auth session management (anon key only in browser)
React Hook Form + Zod	^7.53 / ^3.23	Form state and validation
Recharts	^2.12	Budget breakdown charts
Lucide React	^0.451	Icon library
date-fns	^3.6	Date formatting utilities

Backend

Technology	Version	Purpose
Python	3.12	Runtime (pinned via .python-version)
FastAPI	0.115.0	REST API framework (async)
Uvicorn	0.30.6	ASGI production server
Pydantic / pydantic-settings	2.11.x	Schema validation & env-var loading
Groq SDK	0.11.0	Groq AI API client
Supabase Python	2.33.0	DB read/write via service_role key
PyJWT	>=2.12.0	JWT verification (RS256 / HS256)
Tenacity	9.0.0	Retry with exponential backoff
python-dotenv	1.0.1	Local .env loading
pytest + httpx	8.3.3 / 0.27.2	Test suite (34 tests)

Infrastructure & Services

Service	Tier	Purpose

Supabase	Free (Pro for prod)	PostgreSQL DB + Auth + RLS + JWKS
Groq	Free tier	LLM API – model: openai/gpt-oss-120b
Vercel	Free/Hobby	Frontend hosting + CDN + preview deployments
Render	Free/Starter	Backend hosting – Python web service
GitHub	Free	Source control + CI

Rationale for Key Choices

- **Groq over OpenAI:** Groq's inference hardware delivers <10 s response for 8k-token outputs, critical for a good UX in itinerary generation.
- **Supabase over Firebase:** PostgreSQL + RLS gives structured, queryable storage and true row-level security without custom middleware.
- **FastAPI over Django:** Async-native, lightweight, auto-generates OpenAPI docs (/docs), and starts fast on Render free tier.
- **pydantic-settings:** Provides type-safe environment variable loading with no boilerplate.